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#include <stdio.h>
#define MAX_MEMORY 1000
int memory[MAX_MEMORY];
void initializeMemory()
{
    for(int i = 0; i < MAX_MEMORY; i++)
    {
        memory[i] = -1;
    }
}
void allocateMemory(int processId, int size)
{
    int start = -1;
    int blockSize = 0;
    for(int i = 0; i < MAX_MEMORY; i++)
    {
        if(memory[i] == -1)
        {
            if(blockSize == 0)
            {
                start = i;
            }

            blockSize++;
        }
        else
        {
            blockSize = 0;
        }
        if(blockSize >= size)
        {
            break;
        }
    }
}
```

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    else
    {
        blockSize = 0;
    }
    if(blockSize >= size)
    {
        break;
    }
}
if(blockSize >= size)
{
    for(int i = start; i < start + size; i++)
    {
        memory[i] = processId;
    }
    printf("Allocated Block %d - %d to Process %d\n",
        start,
        start + size - 1,
        processId);
}
else
{
    printf("Allocation Failed\n");
}
}

int main()
{
    initializeMemory();
    allocateMemory(1, 200);
    allocateMemory(2, 300);
    return 0;
}

```

Allocated Block 0 - 199 to Process 1
Allocated Block 200 - 499 to Process 2

Process exited after 0.08122 seconds with return value 0
Press any key to continue . . . |